

Week 16 Capstone Presentation Rubric

CE 310 — Probability & Statistics in Civil and Architectural Engineering

Format: one slide, 2 minutes + 3.5 minutes Q&A (5.5 minutes total per group)

Groups: 4 students per group; each team chose its own dataset — no two teams necessarily analyze the same topic **Total:** 50 points (the slide, and the five and a half minutes you spend on your feet)

Why one slide

The detail is already delivered twice — the notebook holds the analysis and the report explains it to the client. This is the third and shortest form of the same work, and its skill is different from both: choose the one finding that matters and say it in two minutes.

Backup slides are allowed and encouraged — anything you might want on screen while answering a question. They are not presented and not scored as content. Put them after the briefing slide in the same PDF.

The briefing slide

One slide, carrying four things:

Must include	Notes
The finding	The single most important thing your analysis established. Not the method — the finding
The number that supports it	One figure, with units, that your notebook actually produced
The uncertainty around it	Your P10–P90 range, or your confidence interval — one of them, not both
The recommendation	Specific and quantified, addressed to your client

Anything else you show — a figure, a comparison, a table — has to earn its place on the one slide you get.

Scoring Rubric (50 points)

The slide (12 points)

Everything here is fixed before you walk in, and it is the only part of the presentation grade that is.

Criteria	Full	Partial	Minimal	Missing
Finding — one clear claim, stated as a finding rather than a method (4 pts)	4	2–3	1	0
Evidence — a supporting figure with units, traceable to the team's notebook (3 pts)	3	2	1	0
Uncertainty — a range or interval given alongside the point estimate (2 pts)	2	1	—	0
Recommendation — specific and quantified, not generic advice (2 pts)	2	1	—	0
Legibility — readable from the back of the room (1 pt)	1	—	—	0

The five and a half minutes (38 points)

The briefing and the questions are one score, because they are one performance: two minutes to land the finding, then three and a half answering for it, judged live by the same person. **Each of the four members takes one directed question**, and the answers carry the bulk of the 38 — this is the largest single component of the whole capstone, and deliberately the part that cannot be prepared in advance, delegated to a teammate, or produced by a language model.

Delivering it (13 pts)

Criteria	Full	Partial	Minimal	Missing
Leads with the finding — the client knows the answer inside 20 seconds (5 pts)	Finding first, method only if asked	Finding within the first minute	Method first, finding late	Never stated plainly
Plain language — no unexplained statistical jargon (5 pts)	Conversational	Mostly clear	Reads the slide aloud	Incomprehensible
Time — 2 minutes $\pm$ 20 sec (3 pts)	1:40–2:20	1:20–2:40	Outside that range	Stopped at 3:00

Answering for your numbers (25 pts)

Criteria	Full	Partial	Minimal	None
Answers from the team's own results — cites a number the analysis produced (11 pts)	Accurate and specific	Right idea, number vague or wrong	Generic statistics answer	Cannot answer

Criteria	Full	Partial	Minimal	None
Accounts for <i>why</i> the result came out that way, not just what it was — including why an analysis that produced nothing useful produced nothing useful (9 pts)	Explains the mechanism	States the result only	—	—
A member who does not know defers honestly rather than guessing (5 pts)	Says so and offers what they do know	—	Bluffs	Fabricates a number

A team whose numbers are sound but who cannot account for them out loud has not demonstrated the work. Score the answer, not the confidence.

You ran four methods. You are not expected to defend all four as useful. Some of what you ran will have had little to say about your client's question, and saying so is a stronger answer than dressing it up — “the Monte Carlo added nothing here, because our predictor has no independent uncertainty to propagate” demonstrates more judgement than a confident summary of a result that does not mean anything. Knowing which analysis earned its place is the skill this section is scored on, and it is the one that separates an engineer from someone who can run the code.

Every member answers

There is no requirement to split the two-minute briefing three ways — one or two speakers is fine, and forced slide-splitting usually produces three people reading in turn. Instead, **Q&A is directed: one question to each member by name.** That is what guarantees all four speak, and it is far more informative

about who did the work.

AI use

The course policy on AI tools (see the syllabus) applies to the slide as it does to everything else: use it to learn, attribute it if it wrote what you hand in, and own what you submit either way.

The questions are where that matters. You are answering for numbers your team produced, in front of the room, without notes you prepared for that question. A slide drafted by a model you cannot account for is worth 12 points and costs you most of 38.